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LithoGPT-Mini: A Practical Lightweight Language Model for Fast and Accurate Complex Lithology Identification-Application to Underground Gas Storage Facility

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Abstract

Accurate lithology interpretation, full-process automation, and operational efficiency remain difficult to achieve simultaneously in complex reservoir scenarios. This paper focuses on developing a domain-adapted, lightweight generative AI model for intelligent lithology recognition in underground gas storage settings. By using the Wen-23 reservoir—the largest gas storage facility in eastern China—as a case study, the model addresses key challenges in text-based interpretation, thin-layer recognition, and human-in-the-loop decision augmentation.

Wen-23 is the largest gas storage facility in eastern China in terms of working volume and peak-shaving capacity, characterized by highly variable and complex lithology. We introduce LithoGPT-Mini, a compact generative language model based on the open-source LLaMA 3 architecture, optimized via instruction tuning for domain-specific lithology interpretation. The model was trained on a curated corpus of geological texts, including logging data, core descriptions, and expert annotations from Wen-23. It converts unstructured geological descriptions into structured lithology classifications and semantic outputs. The system is further embedded in a human-in-the-loop workflow for rapid expert verification and feedback integration.

LithoGPT-Mini successfully demonstrates high interpretative performance in geologically complex sections, particularly in zones characterized by thin interbeds, variable gamma-resistivity responses, and linguistically ambiguous layer descriptions. The model generates consistent lithology classifications and supports multi-level outputs including depth-wise lithofacies labeling and qualitative descriptions. It adapts well to typical sedimentary features in Wen-23, such as red-bed intervals with high saline depositional signatures, mudstone flooding surfaces, and sandstone interbeds with varying diagenetic textures. When benchmarked against expert-verified interpretations, LithoGPT-Mini achieves a high degree of overlap

in both classification accuracy and semantic correspondence. Compared to rule-based or convolutional approaches, the model offers a more flexible and interpretable mechanism, particularly valuable when dealing with historical datasets lacking structured logs. Additionally, its low computational overhead allows near real-time inference and integration into field workflows. The case study demonstrates the model's capacity to augment geological modeling pipelines and highlights its potential for scalable deployment in gas storage sites and broader reservoir interpretation scenarios. LithoGPT-Mini enhances both the speed and consistency of interpretation, bridging the gap between AI outputs and expert geological understanding.

This work introduces a vertically optimized generative AI solution tailored to lithology recognition, representing one of the first applications of instruction-tuned language models in reservoir geology. By balancing accuracy, automation, and speed, LithoGPT-Mini offers a deployable tool for field-scale geological interpretation. Its implementation in Wen-23 illustrates the feasibility of applying lightweight LLMs to subsurface domains, contributing new strategies for digital geology and energy AI applications.

Introduction

The escalating global demand for energy, coupled with the imperative for energy security and sustainability, has underscored the critical role of underground gas storage (UGS) facilities in balancing supply and demand fluctuations (Abubakar et al. 2022; Ramachandran et al. 2024). These facilities are essential for managing the intermittency of renewable energy sources and ensuring a stable energy supply, particularly given the increasing focus on achieving net-zero emission targets (Epelle et al. 2022; Stephenson et al. 2022). Geological characterisation of UGS sites, especially the accurate identification of lithological units and their petrophysical properties, is crucial to ensure safe, efficient, and long-term operational integrity (Bai et al. 2025; An et al. 2025).

Traditional lithology identification methods, such as core analysis, well logging, and seismic interpretation, though widely used, are often time-consuming, costly, and prone to interpretation biases. These challenges are amplified in complex geological settings, such as evaporite sequences, where heterogeneous lithologies and diagenetic alterations introduce significant uncertainties. In recent years, deep learning techniques have emerged as transformative tools to overcome these limitations by enabling rapid, accurate, and automated lithological classification from subsurface datasets (Misra and Li 2019; Tang et al. 2021; Zhang et al. 2024a). Advanced deep learning models can efficiently process extensive well-log data to predict subsurface lithologies, addressing the shortcomings of traditional methods that often assume spatial independence (Tian et al. 2020). By leveraging machine learning algorithms, it becomes possible to enhance the precision of subsurface characterisation and provide critical insights for reservoir and storage management (Chen et al. 2025; Li et al. 2025).

The rapid evolution of large language models (LLMs), such as GPT-4, has further demonstrated the transformative potential of AI in diverse scientific domains, including geoscience applications (Zhao et al. 2024). Within the petroleum sector, artificial intelligence (AI), encompassing machine learning and deep learning, has shown notable advancements in lithology identification, intelligent fault detection, real-time optimisation of waterflooding operations, and prediction of oil and gas production (Kuang et al. 2021; Sircar et al. 2021; Bai et al. 2025). These applications streamline data processing and interpretation, improving efficiency and decision-making across upstream oil and gas operations. Furthermore, the integration of advanced algorithms, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs), has produced favourable outcomes in geological analysis, log interpretation, and reservoir simulation, significantly enhancing predictive capabilities (Qian et al. 2024; Zhang et al. 2024b). This has contributed to reduced risk factors and maintenance costs, while making interpretation workflows more accessible to a wider range of specialists (Sircar et al. 2021).

Despite these advancements, challenges remain in the widespread adoption of AI technologies in the oil and gas industry, including the need for specialised expertise and the limitations posed by constrained

datasets (Li et al. 2024; An et al. 2025). However, the continuous evolution of AI, coupled with increasing data availability and computational power, indicates that these challenges are surmountable, paving the way for more robust and broadly applicable AI solutions in geological characterisation (Hong et al. 2020; Koroteev and Tekić 2020; Chen et al. 2025). The effectiveness of AI models for subsurface characterisation critically depends on data quality, highlighting the importance of robust data-management strategies to ensure comprehensive training and validation datasets (Li et al. 2025; Sircar et al. 2021).

Recent developments in generative AI and language models promise to further refine analytical capabilities under complex geological conditions, offering new pathways to address challenges that conventional methods cannot readily resolve (Bickley et al. 2024; Li et al. 2024). These technologies enhance information processing and search capabilities, augmenting human expertise in generating novel geological insights (Bickley et al. 2024; Zhao et al. 2024). This facilitates a shift from localised search routines to broader exploration of geological possibilities, thereby boosting productivity and efficiency in subsurface analysis. Consequently, the integration of AI not only optimises existing workflows but also deepens the understanding of complex geological systems by identifying subtle patterns often overlooked by traditional approaches (Zhang et al. 2024b; Koroteev and Tekić 2020).

In this context, the present study introduces LithoGPT-mini, a lightweight generative language model built on the LLaMA-3.2-1B architecture. The study explicitly explores end-to-end lithology identification using a pure language model approach, without external tool augmentation, aiming to examine whether a compact 1B-scale model can be effectively adapted for geological interpretation tasks. To this end, we design a complete workflow covering data collection, preprocessing, fine-tuning, inference, and human-in-the-loop verification, thereby establishing a reproducible pipeline for geological characterisation. Given its compact scale, LithoGPT-mini can be efficiently deployed on resource-constrained or edge devices, supporting near real-time applications in underground gas storage operations. To evaluate model adaptability, two fine-tuning strategies are systematically compared: Low-Rank Adaptation (LoRA) for parameter-efficient optimisation and full-parameter fine-tuning for maximal accuracy. This comparative study provides new insights into the trade-offs between efficiency and predictive reliability, and underscores the potential of compact LLMs for scalable and automated lithology interpretation in critical energy infrastructure.

Geological Setting and Data

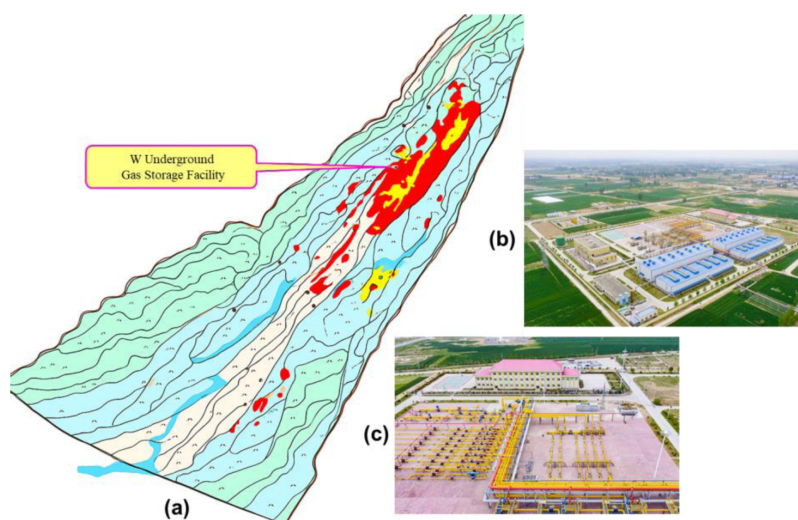


Figure 1—Location and surface facilities of the W Underground Gas Storage Facility: (a) Structural map of the W Underground Gas Storage Facility area showing the distribution of the storage formation (red) and surrounding geological features; (b) Panoramic view of the surface gas storage facility; (c) Layout of the injection and production pipeline system.

Overview of the W UGS Facility

The W Underground Gas Storage (UGS) facility is located in the Wendeng Depression of the Bohai Bay Basin. It represents a typical UGS site developed through the modification of thick evaporite sequences. Since the Paleogene, prolonged evaporative conditions have led to the deposition of massive halite layers interbedded with various evaporite facies, including glauberite-bearing rock salt, argillaceous rock salt, and saline mudstone. Thin interbeds of mudstone and gypsum-bearing mudstone are also common within the halite, showing distinct contrasts in ductility and mechanical behaviour.

After modification, natural gas is primarily stored within dissolution cavities and microfracture systems of the high-porosity, low-permeability salt formations. The overlying mudstone and gypsum-bearing mudstone act as effective cap rocks, providing excellent sealing capacity. Structurally, the area is characterised by a gently dipping syncline with limited fault development, which further enhances cap-rock stability.

The depositional setting reflects a transition from marginal shallow-lake to evaporative-lagoon environments, resulting in significant vertical lithofacies alternations. These include thick halite units interbedded with thin mudstone and argillaceous rock salt, locally enriched with glauberite-bearing halite. The extremely low permeability and high ductility of halite are the key factors ensuring the long-term safety of underground gas storage.

Dataset Construction

The studied interval extends from 970 m to 1116 m and comprises multiple evaporitic and clastic facies. As shown in Fig. 2, the GR, DEN, AC, and RES logs display distinct responses to different lithologies, while the stacked bar chart illustrates vertical variations in clay, halite, glauberite, and gypsum contents. Eight representative lithologic sections were selected to capture the diversity of rock types and were annotated in the log display for reference.

To build a reliable dataset, lithology labels were determined through a combination of manual interpretation of well logs and calibration against core descriptions. This ensured that the dataset retained both quantitative petrophysical responses and qualitative geological reasoning. A total of 94 well intervals were collected from multiple wells within the W Underground Gas Storage Facility, covering lithologies such as rock salt, rock salt with glauberite, salt rock with mudstone, mudstone with salt, argillaceous rock salt, saline mudstone, mudstone with gypsum, and mudstone. These lithologies exhibit pronounced vertical and lateral heterogeneity, creating significant challenges for automated machine-learning models.

For model training, all well-log measurements were converted into structured natural-language prompts. Numerical values such as depth, gamma-ray (GR), density (DEN), acoustic slowness (AC), and resistivity (RES) were formatted into descriptive text strings, while geological reasoning was encoded in the responses. The dataset was prepared in a ChatML instruction–response format, where the system defines the task, the user provides the input (numerical logs), and the assistant generates a geologically informed interpretation. A typical example is shown below:

```
{
  "messages": [
    {
      "role": "system",
      "content": "You are an expert geologist. Analyze well logs and provide lithology interpretation with reasoning, depositional environment, and sealing/reservoir characteristics."
    },
    {
      "role": "user",
      "content": "Depth: 983.5 m; GR: 22.5 API; DEN: 2.13 g/cm³; AC: 220.9 µs/m; RES: 55749.7 Ω·m."
    },
    {
      "role": "assistant",
      "content": "The curves show extremely low gamma-ray and neutron responses, with very high resistivity, indicating halite as the main mineral. Deposited in an extremely strong evaporitic environment, this interval has excellent sealing. The lithology is Rock Salt.<end>"}
  ]
}
```


This design provides explainable training samples, enabling the model not only to classify lithologies but also to learn domain-specific geological reasoning such as depositional environment interpretation and sealing/reservoir characterisation.

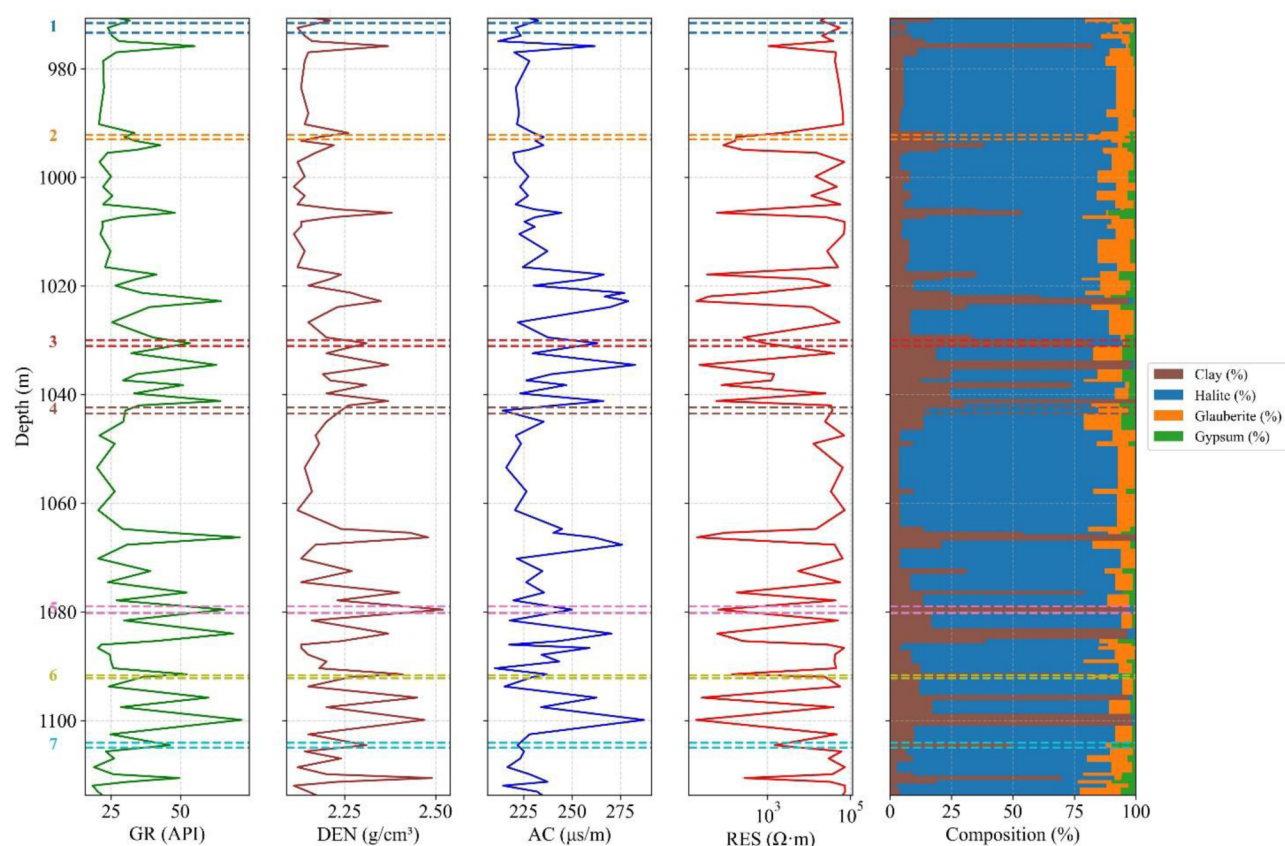


Figure 2—Well log display of J well in the W Underground Gas Storage Facility gas storage (970–1116 m), showing: (a) GR, DEN, AC, and RES curves; (b) stacked mineral composition (Clay, Halite, Glauberite, Gypsum); (c) representative lithology intervals (1–8).

For fine-tuning the proposed LithoGPT-mini model, all well-log measurements were reformatted into natural-language instruction–response pairs. Specifically, numerical logging values such as depth, gamma ray (GR), density (DEN), acoustic slowness (AC), and resistivity (RES) were converted into descriptive text strings, ensuring consistency across samples and enabling the model to directly process structured geological information in a language-like format. Each training instance was framed as a geoscientific dialogue, in which the system defined the task, the user provided well-log inputs, and the assistant produced an expert-level interpretation.

The assistant responses were designed to be explainable and context-aware, going beyond simple lithology classification to include reasoning about geophysical curve responses, depositional environments, and reservoir or sealing properties. This approach allowed the model to capture not only lithological categories but also the underlying geological rationale, thereby supporting interpretable AI-assisted workflows.

Representative input–output pairs are summarised in Table 1, which illustrates how raw well-log data were transformed into natural-language queries and how the model was trained to produce structured, geologically informed outputs. This design ensures that LithoGPT-mini learns to perform end-to-end lithology interpretation, while retaining transparency and adaptability in complex subsurface scenarios.

Table 1—Sample queries sent to LithoGPT-mini and its responses, .

No.	System	User (Well Log Input)	Assistant (Interpretation)
1	You are an expert geologist. Analyze well logs and provide lithology interpretation with reasoning, depositional environment, and sealing/reservoir characteristics.	Depth: 970.4 m; GR: 23.9 API; DEN: 2.13 g/cm ³ ; AC: 223.7 μ s/m; RES: 29584.8 Ω ·m.	The log responses show very low gamma-ray and neutron, high resistivity, indicating salt minerals as the main component with extremely low porosity. Formed in a strong evaporitic environment, it has superior sealing. The lithology is Rock Salt.
2	Same as above	Depth: 969.4 m; GR: 25.0 API; DEN: 2.19 g/cm ³ ; AC: 221.1 μ s/m; RES: 49474.2 Ω ·m.	High resistivity and low density, sonic curves showing higher hardness, and neutron indicating low porosity; low gamma-ray reflects salt minerals dominance with minor sulfate minerals. Deposited in a strong evaporitic environment, it has excellent sealing. The lithology is Rock Salt with Glauberite.
3	Same as above	Depth: 892.2 m; GR: 85.0 API; DEN: 2.48 g/cm ³ ; AC: 301.1 μ s/m; RES: 1.2 Ω ·m.	The curves still show high gamma-ray, indicating clay dominance, but sonic and neutron logs suggest local hard mineral infill and slightly higher density. Gypsum mineral deposition during sedimentation improved rock mechanical properties. The lithology is Mudstone with Gypsum.
4	Same as above	Depth: 975.8 m; GR: 55.1 API; DEN: 2.37 g/cm ³ ; AC: 261.8 μ s/m; RES: 1067.9 Ω ·m.	Logs show high gamma-ray, sonic and neutron indicating certain porosity, and increased salt content causing density variation. The depositional environment represents mud deposition mixed with salt, forming a dense caprock. The lithology is Mudstone with Salt.
5	Same as above	Depth: 971.2 m; GR: 31.8 API; DEN: 2.21 g/cm ³ ; AC: 232.5 μ s/m; RES: 19236.2 Ω ·m.	The curves show gamma-ray slightly higher than pure salt layers, indicating some clay within the salt rock, resulting in slightly increased porosity. Deposited in an evaporitic environment with external clay input, sealing remains good. The lithology is Salt Rock with Mudstone.
6	Same as above	Depth: 971.2 m; GR: 31.8 API; DEN: 2.21 g/cm ³ ; AC: 232.5 μ s/m; RES: 19236.2 Ω ·m	Logs display variable resistivity and gamma-ray responses, reflecting alternating layers of mud and evaporite minerals. This points to Salt Rock with Mudstone, deposited under fluctuating salinity conditions. The heterogeneity results in variable sealing and limited reservoir potential.
7	Same as above	Depth: 1042.0 m; GR: 35.1 API; DEN: 2.26 g/cm ³ ; AC: 241.7 μ s/m; RES: 34368.4 Ω ·m.	Logging curves show high clay content with some salt, resistivity moderate. Deposited during a stage of enhanced mud input at the lake margin. The lithology is Argillaceous Rock Salt.
8	Same as above	Depth: 1104.5 m; GR: 46.2 API; DEN: 2.31 g/cm ³ ; AC: 221.7 μ s/m; RES: 1488.5 Ω ·m.	High gamma-ray and sonic responses show clay dominance with minor salt, resistivity slightly higher than typical mudstone. Deposited in a mixed zone at the edge of an evaporitic lake. The lithology is Saline Mudstone.

Notes: GR = Gamma ray (API); DEN = Density (g/cm³); AC = Acoustic slowness (μ s/m); RES = Resistivity (Ω ·m).

Methodology

Model Framework Overview

The backbone of the proposed LithoGPT-Mini is the LLaMA 3.2–1B architecture, which adopts a decoder-only transformer design optimised for generative tasks. As illustrated in Fig. 3a, the model consists of an embedding layer, multiple stacked transformer blocks, and a final projection with softmax output. Input tokens are mapped into high-dimensional vectors through token embeddings, while sequential order is represented using Rotary Position Embeddings (RoPE), enabling effective modelling of long well-log sequences.

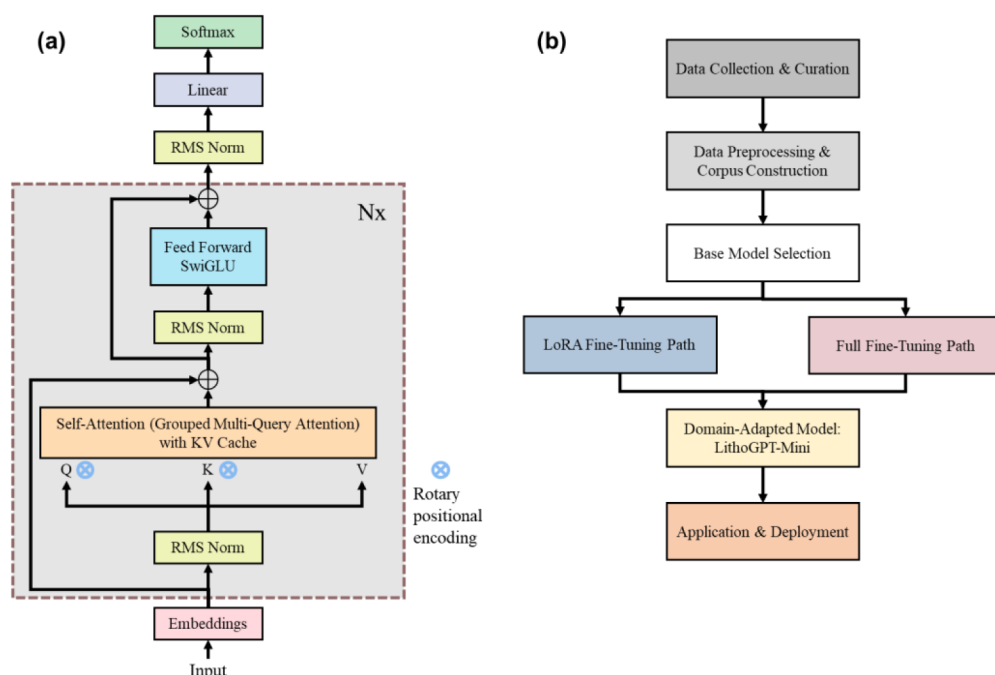


Figure 3—Framework of the proposed LithoGPT-Mini model: (a) Architecture of the LLaMA 3.2-1B transformer; (b) Workflow from data preparation to fine-tuning and deployment.

Each transformer block integrates two principal modules, both preceded by Root Mean Square Normalisation (RMSNorm) to stabilise training and improve efficiency. The first module is a self-attention mechanism with Grouped Multi-Query Attention (GQA), where multiple query heads share a smaller set of key-value projections. This design reduces memory and computational overhead while maintaining attention performance. A KV cache further accelerates inference by reusing previously computed key-value pairs, which is particularly beneficial for near real-time deployment. The second module is a feed-forward network (FFN) with SwiGLU activation, an enhanced gated function that improves representational capacity compared with conventional GeLU or ReLU. Residual connections around both modules facilitate gradient flow and stable optimisation.

The final hidden states are projected through a linear layer and normalised via softmax to produce probability distributions. In lithology interpretation tasks, these outputs correspond to categories and descriptors such as lithology type, depositional environment, and reservoir or sealing properties. The combination of RoPE, GQA, SwiGLU, and RMSNorm improves efficiency, accuracy, and stability, making the LLaMA 3.2-1B model a robust foundation for parameter-efficient fine-tuning in underground gas storage lithology interpretation.

Workflow

The complete workflow of the proposed LithoGPT-Mini framework is illustrated in Fig. 3b. The process begins with data collection and curation, where well-log measurements, core descriptions, and expert annotations from the W Underground Gas Storage Facility are compiled into a unified corpus.

The data undergoes preprocessing and corpus construction, including text cleaning, annotation alignment, and formatting into instruction-response pairs. Well-log curves are transformed into natural-language strings, while geological reasoning is encoded in structured outputs, enabling the model to learn both quantitative curve responses and qualitative domain semantics.

The base model selection stage uses the LLaMA 3.2-1B architecture, followed by two parallel fine-tuning strategies: Low-Rank Adaptation (LoRA) fine-tuning, which introduces lightweight adaptation layers, and full fine-tuning, which updates all model weights at higher computational cost. Both approaches

yield a domain-adapted model, LithoGPT-Mini, which is embedded into a human-in-the-loop verification workflow for rapid expert validation, feedback integration, and iterative refinement.

Finally, the adapted model is deployed in application workflows, enabling automated lithology interpretation, thin-layer recognition, and semantic description generation. Its compact architecture allows deployment on resource-constrained environments, supporting near real-time interpretation in underground gas storage operations while remaining extensible to broader reservoir characterisation tasks.

Training and Optimization

To adapt the pre-trained LLaMA 3.2–1B model to the lithology interpretation task in the W Underground Gas Storage Facility, we fine-tuned a customised model, LithoGPT-mini, using domain-specific Chat Markup Language (ChatML) formatted datasets. Two training strategies were evaluated: Low-Rank Adaptation (LoRA) and full fine-tuning. Both strategies were run under identical maximum training steps to ensure a fair comparison.

For the LoRA-based fine-tuning, only the low-rank adaptation layers were updated while freezing backbone parameters. This approach significantly reduced computational cost and memory footprint. In contrast, full fine-tuning updated all parameters of the base model, providing greater flexibility in parameter adjustment at the expense of higher computational demand.

The optimisation process combined a cross-entropy loss with a domain-guided regularisation term. This encouraged the model to capture lithology-related semantics while reducing overfitting to noise in the training set. The `paged_adamw_32bit` optimiser was applied in the LoRA configuration to efficiently handle quantised parameters, while `adamw_torch` was employed in full fine-tuning.

Hyperparameters such as batch size, gradient accumulation steps, and learning rate were tuned to balance convergence speed and generalisation ability. The training configurations for both strategies are summarised in Table 2, showing distinct differences in quantisation precision, number of trainable parameters, and optimisation hyperparameters. This controlled design enables a clear evaluation of the trade-offs between lightweight parameter-efficient fine-tuning and full model adaptation in terms of accuracy and computational efficiency.

Table 2—Training parameters for LoRA-based and full-parameter fine-tuning experiments.

Parameter	LoRA Fine-tuning	Full-parameter Fine-tuning
Base Model	LLaMA 3.2–1B	LLaMA 3.2–1B
Quantization	4-bit NF4	None (16-bit FP)
Trainable Parameters	Low-rank adaptation layers	All parameters
Batch Size (per device)	6	2
Gradient Accumulation Steps	4	8
Learning Rate	2×10^{-4}	2×10^{-5}
Weight Decay	0.0	0.01
Warmup Steps	20	20
Max Training Steps	200	200
Loss Function	Cross-entropy	Cross-entropy
Optimizer	<code>paged_adamw_8bit</code>	<code>adamw_torch</code>

Results and Discussions

Training Performance and Computational Efficiency

To evaluate the effectiveness of the two fine-tuning strategies, both Low-Rank Adaptation (LoRA) and full fine-tuning were applied to the LLaMA 3.2–1B model using the same dataset and training configuration. Fig. 4 shows the evolution of training loss and the corresponding training time for both approaches. The loss curves exhibit a similar trend, with rapid convergence within the first 25 steps under the selected hyperparameters. Although LoRA displays slightly slower reduction in the early stage, both strategies ultimately achieve comparable final loss levels.

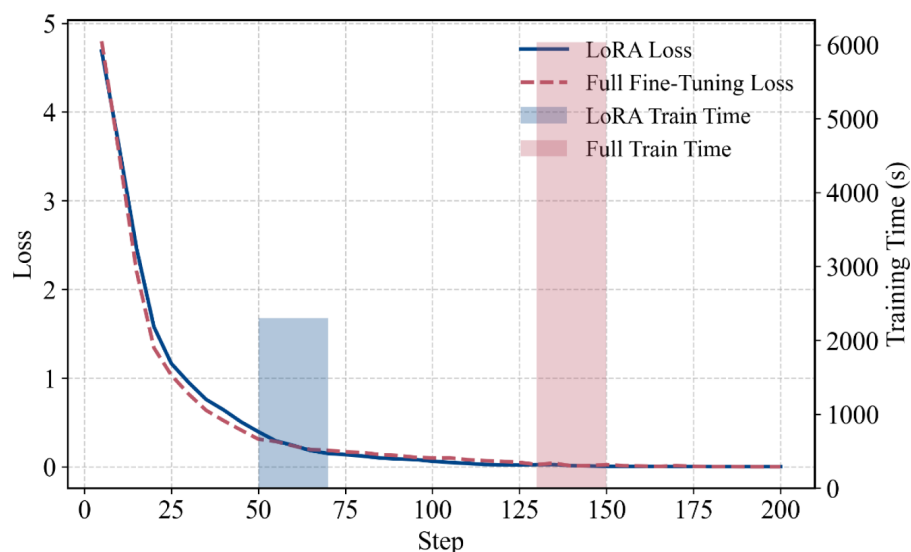


Figure 4—Training loss curves and training time comparison between LoRA-based fine-tuning and full-parameter fine-tuning.

The recorded training times highlight a clear difference in computational efficiency. The blue band indicates that LoRA training completed in less than 2000 s, whereas the red band shows that full fine-tuning required over 6000 s. These results confirm that LoRA provides a substantial reduction in training cost while maintaining accuracy comparable to full fine-tuning.

Comparison of Lithology Prediction Accuracy and Semantic Consistency

The bar chart in Fig. 5 compares the performance of LoRA and full fine-tuning in terms of lithology prediction accuracy and mean semantic similarity. A clear performance gap is observed in lithology prediction accuracy, where full fine-tuning achieves 0.989, while LoRA reaches only 0.426. In contrast, both methods show comparable results in mean semantic similarity, with scores of 0.811 for LoRA and 0.986 for full fine-tuning. These results suggest that although LoRA is capable of producing semantically consistent outputs, it struggles to accurately classify lithology types compared with the full-parameter model.

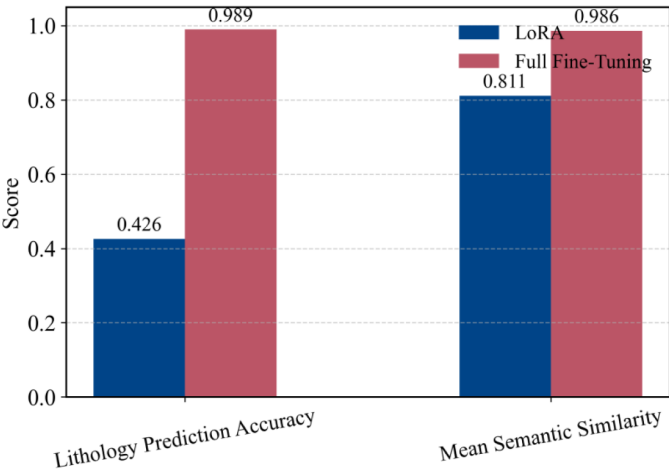


Figure 5—Comparison of lithology prediction accuracy and mean semantic similarity between LoRA-based and full-parameter fine-tuning.

Confusion Matrix and Semantic Similarity Distribution

The confusion matrices in Fig. 6 provide a detailed comparison of classification performance across eight lithology types. The full fine-tuning model shows clear diagonal dominance, with most samples correctly classified into their true lithology classes. High accuracies are evident for rock salt (24/25) and rock salt with glauberite (21/22), as well as consistent recognition of salt rock with mudstone and argillaceous rock salt. In contrast, the LoRA model displays higher off-diagonal values, reflecting frequent misclassifications among salt-bearing lithologies. For example, instances of rock salt are often confused with rock salt with glauberite, and salt rock with mudstone is inconsistently separated from mudstone with salt. This indicates that LoRA struggles to capture the subtle geophysical distinctions between lithologies with similar evaporitic compositions. These results are consistent with the overall accuracy evaluation in Fig. 5, confirming that LoRA is less precise in lithology classification compared with full fine-tuning.

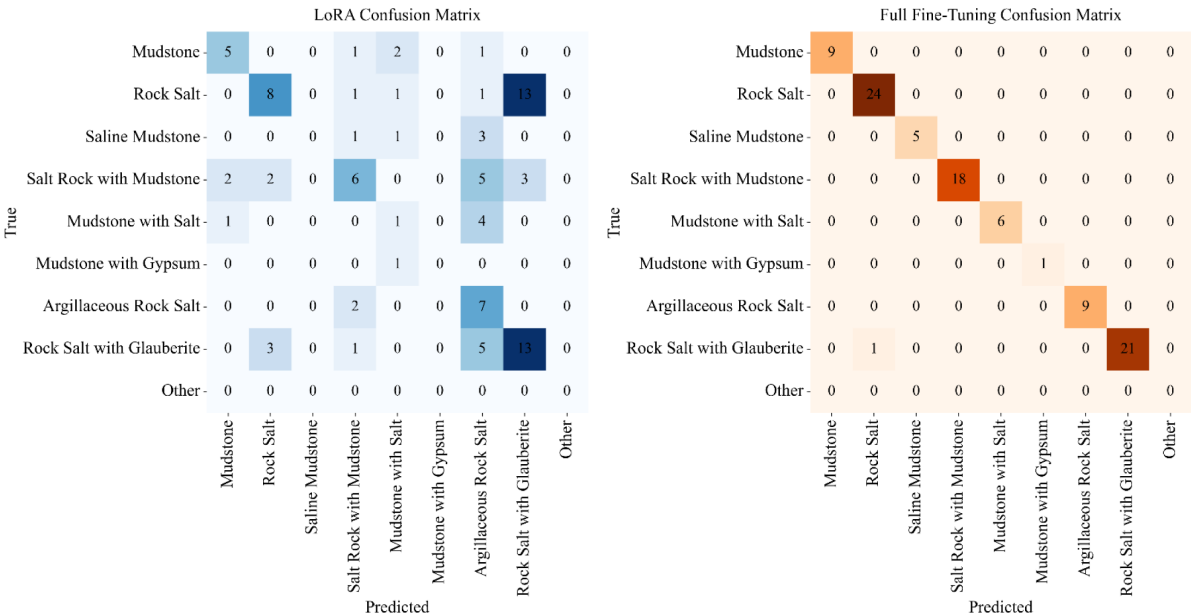


Figure 6—Confusion matrices of LoRA-based fine-tuning (left) and full-parameter fine-tuning (right) for lithology classification.

The semantic similarity distributions in Fig. 7 illustrate how well model outputs align with geologically meaningful text descriptions. Both models achieve similarity scores concentrated towards high values,

indicating that generated interpretations remain geologically coherent. However, full fine-tuning produces a sharper distribution peaking near 1.0, with fewer low-similarity cases, suggesting more consistent semantic alignment. By contrast, the LoRA model exhibits a broader distribution, with outputs ranging from 0.6 to 1.0, reflecting greater variability in interpretative quality. Despite this dispersion, LoRA still maintains a reasonable similarity centre around 0.8, showing that it is capable of capturing essential geological semantics even when exact lithology predictions are incorrect. These findings highlight the distinction between classification accuracy and semantic consistency, and demonstrate that lightweight fine-tuning strategies can preserve domain semantics while compromising precision in categorical prediction.

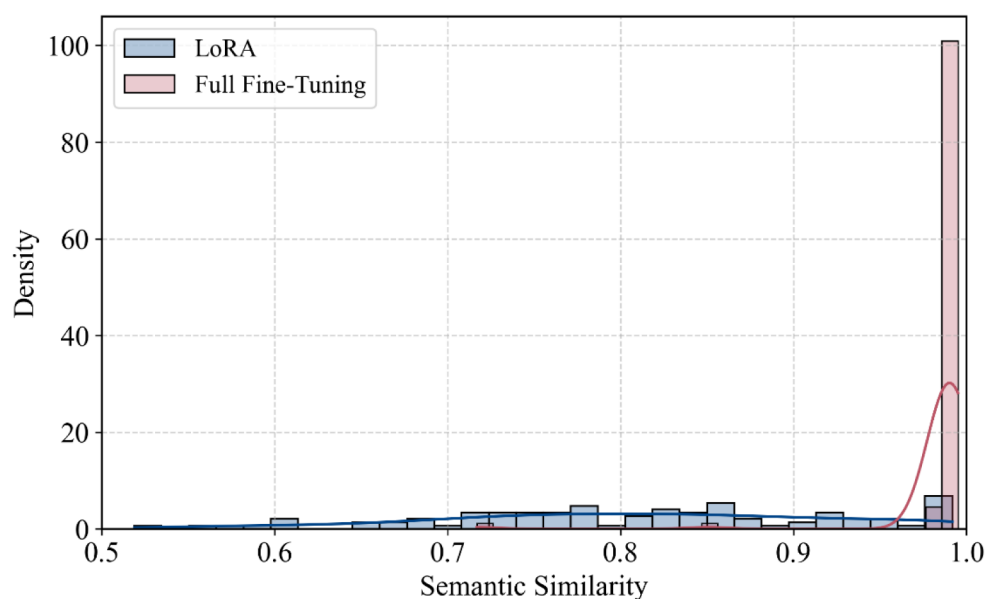


Figure 7—Semantic similarity distributions of predictions from LoRA and full-parameter fine-tuning.

Case Study

To further evaluate the practical interpretability of the two fine-tuning strategies, three representative cases were selected from the test dataset, covering lithologies with distinct geological characteristics: Rock Salt, Rock Salt with Glauberite, and Mudstone with Gypsum. The model outputs for these cases are shown in Fig. 8, where predictions from both LoRA and full fine-tuning are compared side by side.

For the Rock Salt case, both models correctly identified the lithology as rock salt. The full fine-tuning model produced concise descriptions that directly emphasised the evaporitic depositional setting and superior sealing capacity of halite formations. In contrast, the LoRA model also captured the evaporitic environment but frequently introduced additional mixed-lithology elements in its text, resulting in less precise classification.

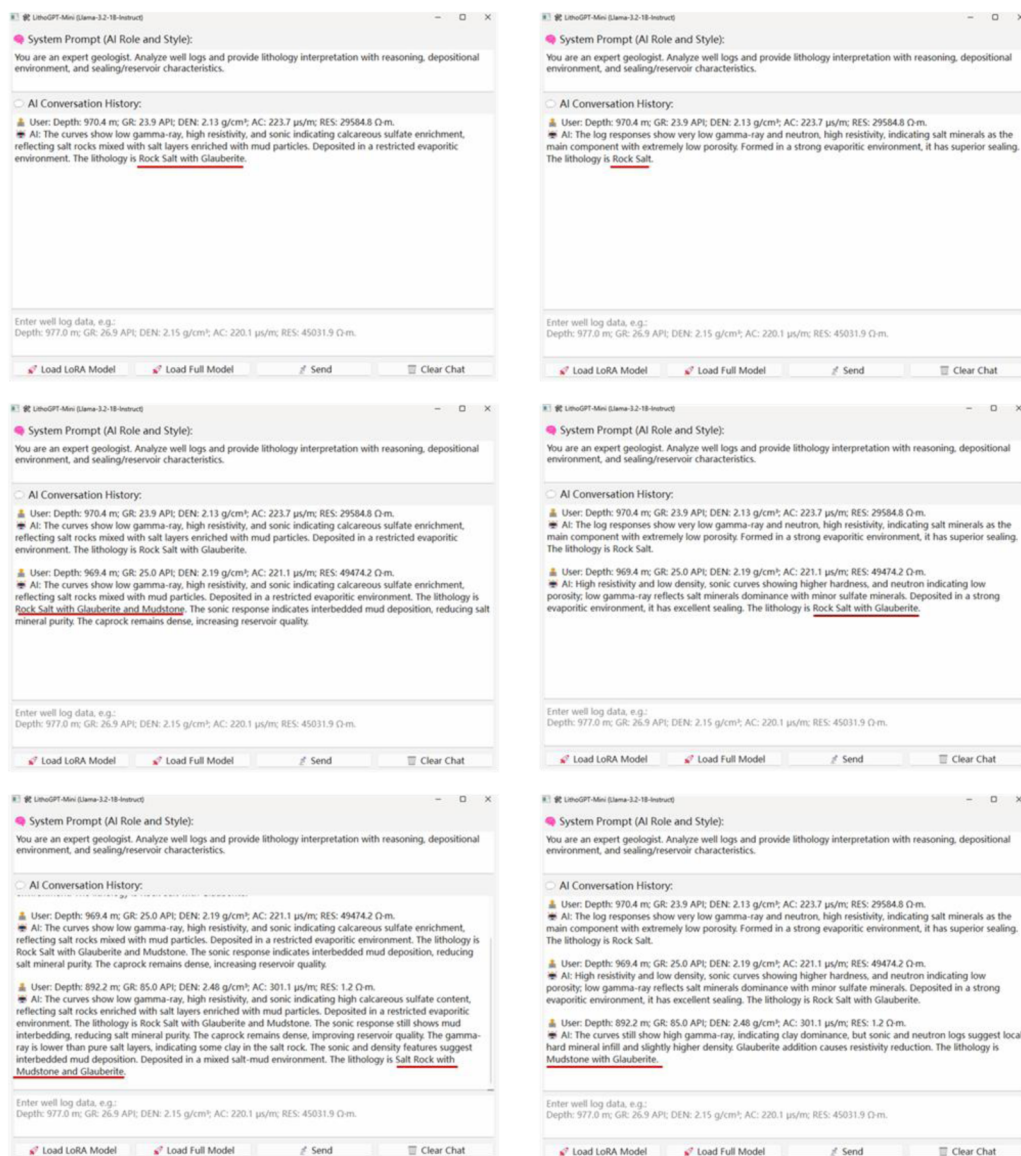


Figure 8—Comparison of lithology interpretation outputs from LoRA-based and full-parameter fine-tuning models on three representative test samples.

In the case of Rock Salt with Glauberite, differences between the two approaches became more evident. The full fine-tuning model consistently recognised the presence of glauberite and explicitly described its mineralogical effect on rock mechanical behaviour. By comparison, the LoRA model misclassified this interval as pure rock salt in several instances, with only vague references to mineral heterogeneity. This observation aligns with the confusion matrix analysis in Fig. 6, which indicated that LoRA struggles to differentiate between similar evaporitic lithologies.

For Mudstone with Gypsum, both models correctly interpreted the interval as mudstone-dominated, consistent with the high gamma-ray response. However, the full fine-tuning model explicitly identified the gypsum content and discussed its role in improving rock strength and mechanical stability. The LoRA output, on the other hand, generated more generalised descriptions that overlooked the gypsum contribution, leading to incomplete lithology characterisation.

These case studies confirm that while LoRA can generate semantically coherent geological descriptions, it lacks the precision in lithology differentiation achieved by full-parameter fine-tuning.

Conclusion

This study developed LithoGPT-mini, a lightweight language model based on the LLaMA-3.2-1B architecture for lithology interpretation in the W Underground Gas Storage Facility, the largest of its kind in eastern China. The model represents an exploratory attempt to achieve lithology identification using language modelling only, without external tool augmentation. A complete workflow was constructed, covering data collection, preprocessing into ChatML instruction–response pairs, fine-tuning, evaluation, and deployment, showing that even a compact 1B-scale LLM can perform end-to-end lithology interpretation when trained on domain-specific datasets.

Two fine-tuning strategies were compared: LoRA and full parameter fine-tuning. Both achieved similar convergence, but the full model delivered higher classification accuracy, correctly distinguishing subtle variations such as rock salt vs. rock salt with glauberite. LoRA exhibited lower accuracy, especially for salt-bearing lithologies, but retained comparable semantic consistency and required far fewer resources, making it attractive for deployment in constrained environments. Confusion matrix analysis and case studies confirmed these patterns, and semantic similarity evaluation showed that LoRA preserved meaningful geological reasoning even when misclassifications occurred.

Overall, LithoGPT-mini demonstrates the potential of lightweight LLMs in geological characterisation of underground gas storage and similar reservoir settings. The study clarifies both abilities and limitations of compact language models: LoRA offers efficiency and scalability, while full fine-tuning provides superior predictive reliability for complex lithological systems.

However, several limitations remain. The dataset used for training was relatively small, both in sample size and in lithological diversity, which restricts the generalisability of the results. Only a limited number of lithology categories were included, meaning that the model's ability to discriminate a broader spectrum of rock types has yet to be tested. In addition, LithoGPT-mini was built upon the LLaMA-3.2-1B architecture, a compact model that, while efficient, inherently constrains representational capacity compared with larger-scale LLMs. These factors together suggest that the findings should be viewed as exploratory, and further work is required with larger datasets, more diverse lithological classes, and potentially larger models to validate and extend the conclusions.

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